

The Impact of Objective and Perceived Environmental Deprivation on Adolescent Hippocampal Myelination

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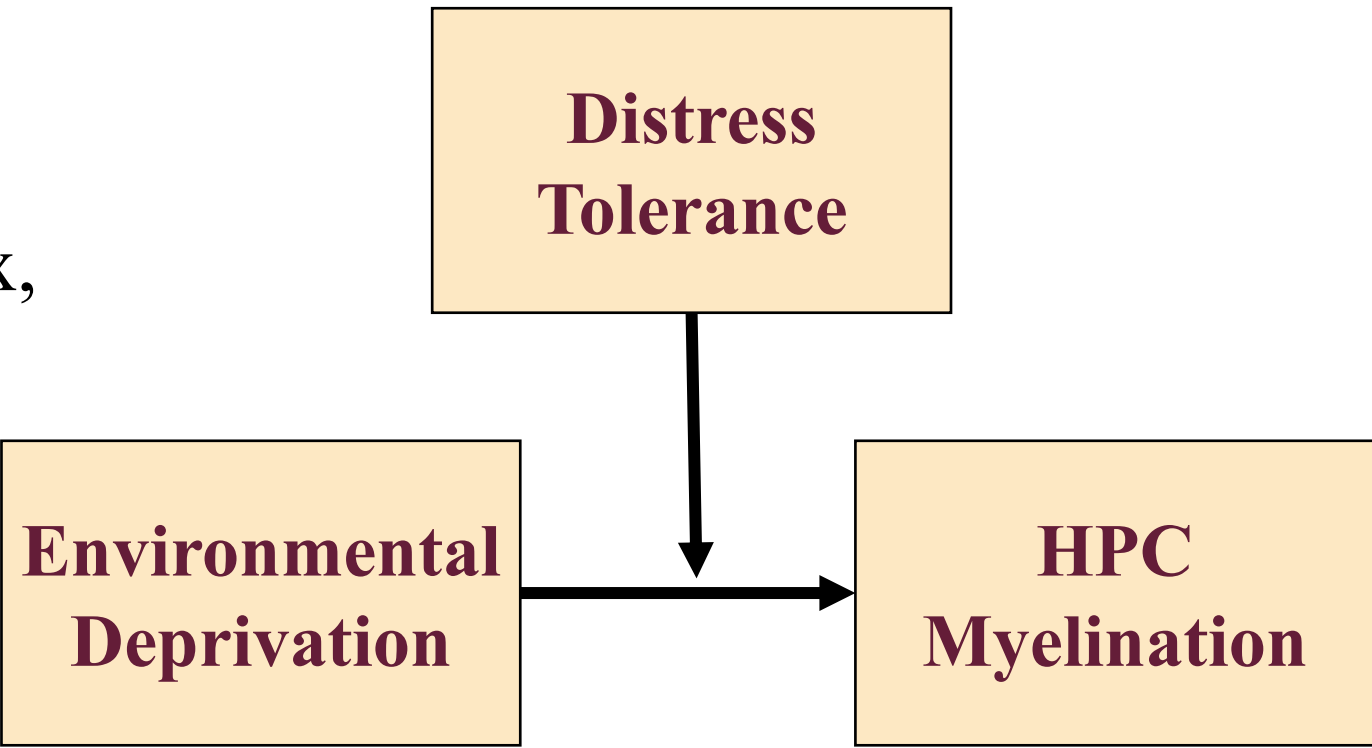
Background

- The hippocampus (HPC) is critical for environmental adaptation as its structure and microstructure continue to mature through adolescence reflecting unique plasticity.^{1, 2}
- Previous research has shown an association between smaller HPC volumes and environmental deprivation.^{2, 3, 4}
- Animal studies suggest that prolonged HPC myelination may underlie responsiveness to environmental impact. However, limited work has examined how HPC myelination unfolds throughout human adolescence and how environmental factors shape myelination trajectories.^{5, 6}
- Distress Tolerance, how individuals respond to negative emotions, may impact how one interacts with their environment and may be shaped by early life environment. Yet, its role in adaptability and relationship to HPC development remain relatively unexplored.^{7, 8}

To bridge these gaps, this study aimed to use 7T quantitative MRI to:

- Examine how HPC myelination is refined from adolescence into young adulthood.
- Examine how objective and perceived environmental deprivation are associated with markers of HPC myelination.
- Examine how greater Distress Tolerance – a marker of adaptive coping – may buffer against deprivation effects on HPC myelination

Hypothesis: We hypothesize that HPC myelination will increase with age, consistent with age-related increases in myelin across the cortex, but will be blunted by objective deprivation and poorly perceived environmental quality. We additionally predict that greater distress tolerance will act as a buffering factor against these effects.



Methods

Participants

140 participants (baseline ages 10-32; 50% Female) were scanned up to 3 times at 18-month intervals (total 224 visits).

Hippocampal Myelination

- Ultra-high field quantitative (7T) MRI scans performed on a Siemens Magnetom. MP2RAGE-derived R1 was extracted from HPC with FreeSurfer segmentation.

Neighborhood Deprivation Index

- Objective neighborhood deprivation was indexed using the **Area Deprivation Index (ADI)** with higher scores reflecting greater deprivation. Perceived environmental quality was assessed using **WHO's Quality of Life Environmental (WHOQOL-ENV)** domain, with lower scores indicating greater perceived deprivation.

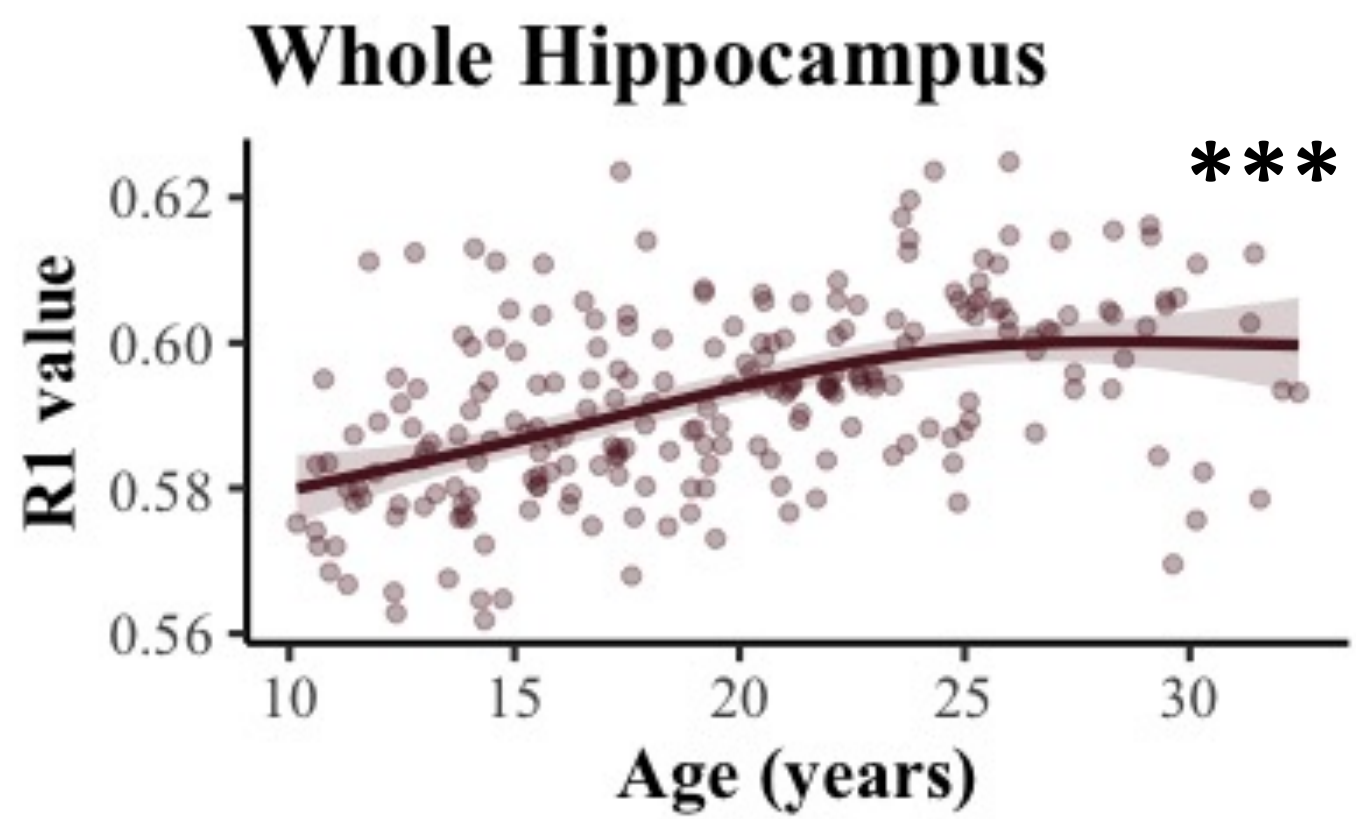
Distress Tolerance

- Distress tolerance was assessed using a 15-item self-report **Distress Tolerance Scale (DTS)** and was categorized into four subscales: Tolerance, Absorption, Appraisal, and Regulation. Higher scores reflect greater inability to tolerate forms of distress.

Statistical Analysis

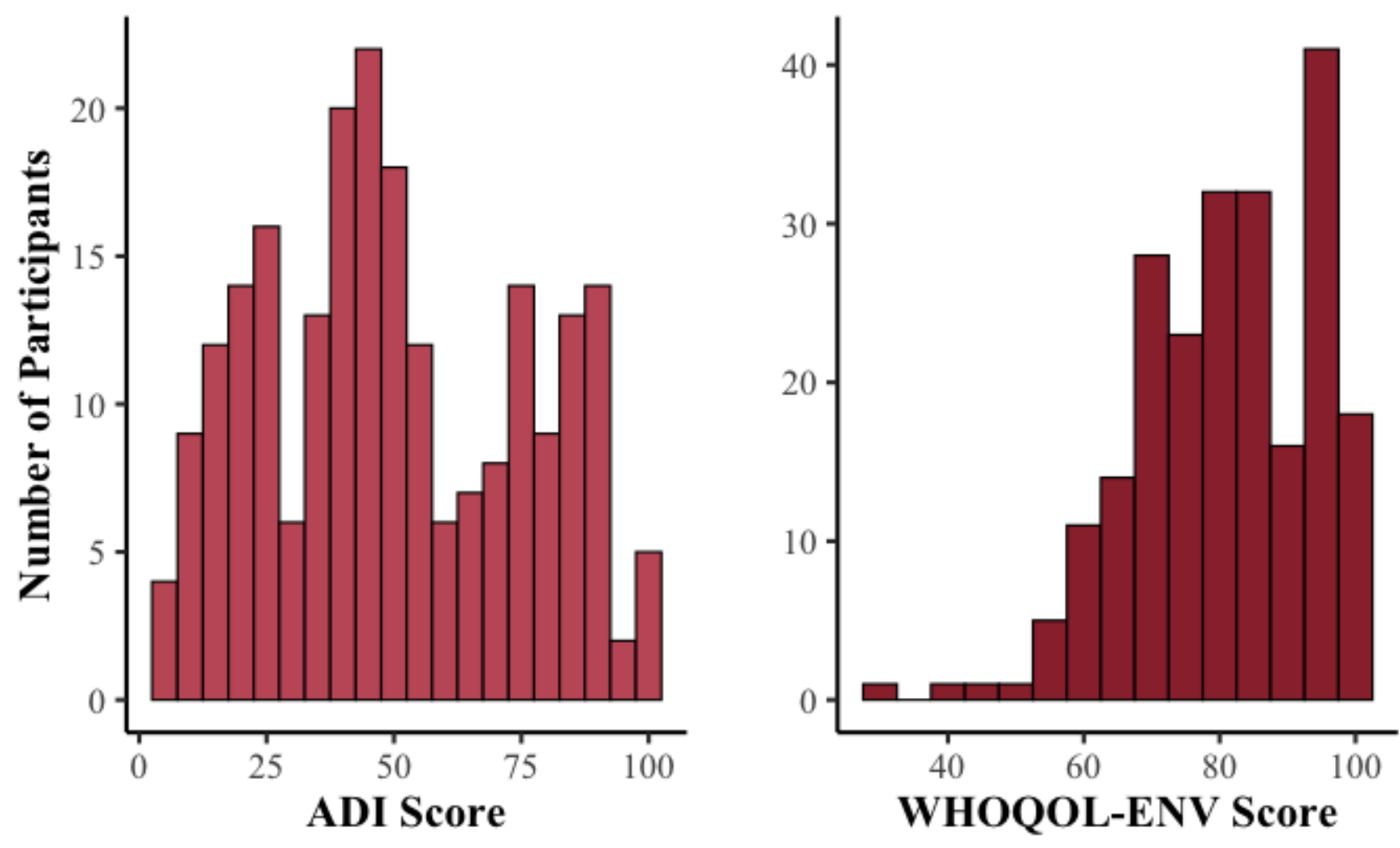
- Generalized additive mixed models (GAMMs) tested nonlinear age effects and associations between R1 and environmental variables.

R1 increases non-linearly within the adolescent hippocampus

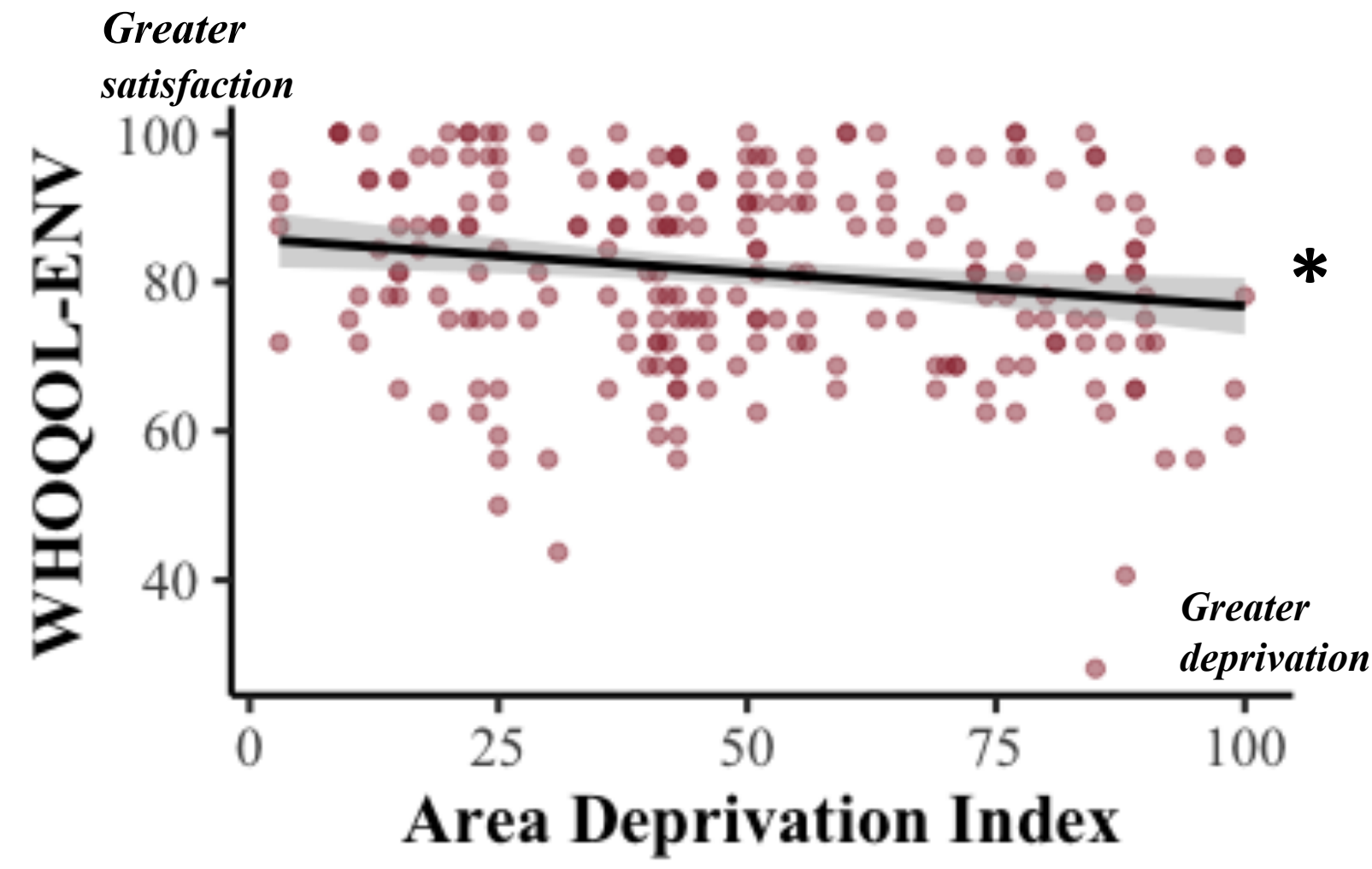
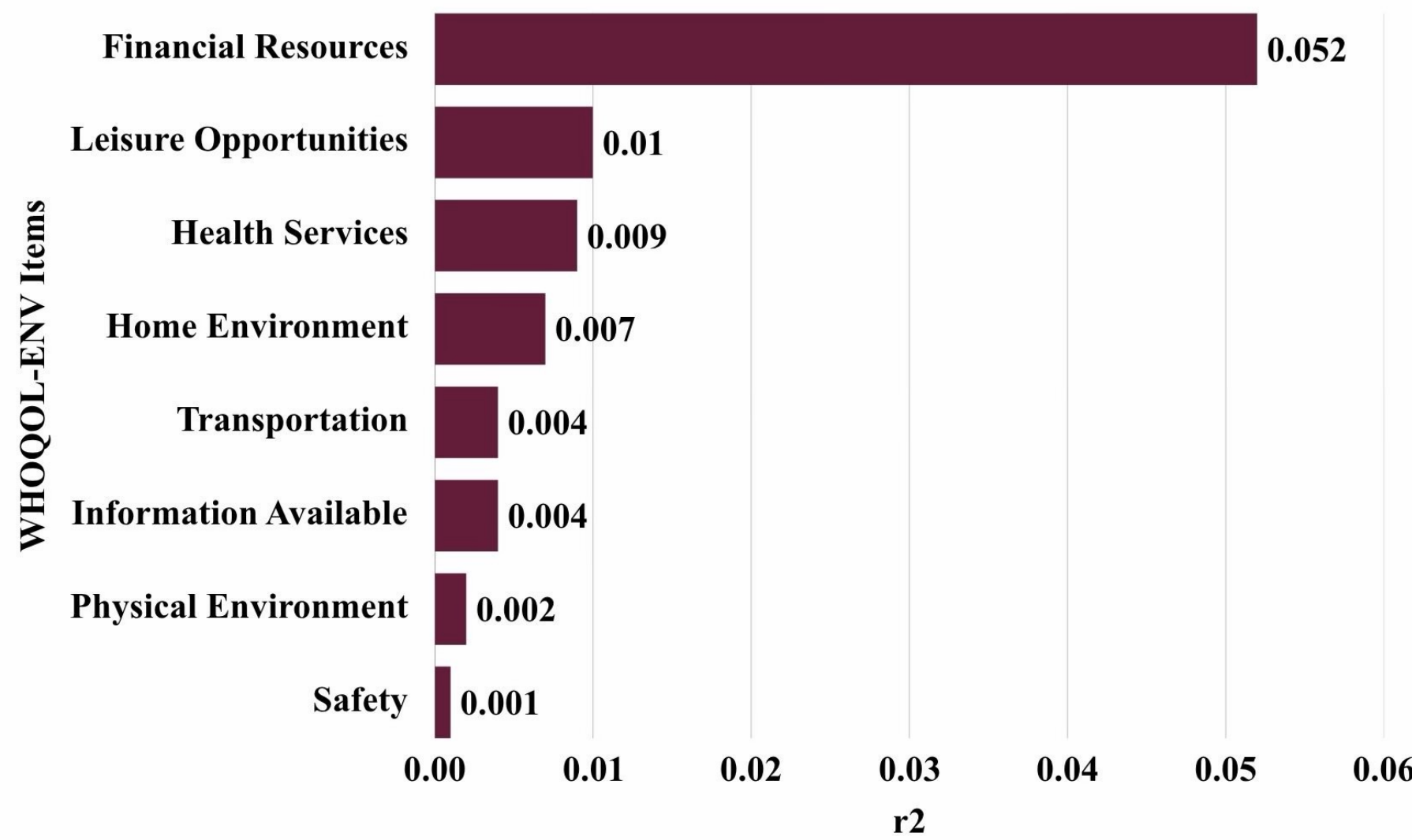


HPC R1 showed significant nonlinear age-related increases (whole and both hemispheres; $p < 0.0001$).

Area Deprivation Index and WHO Quality of Life Environmental measures capture both shared and unique variance in youths' environments



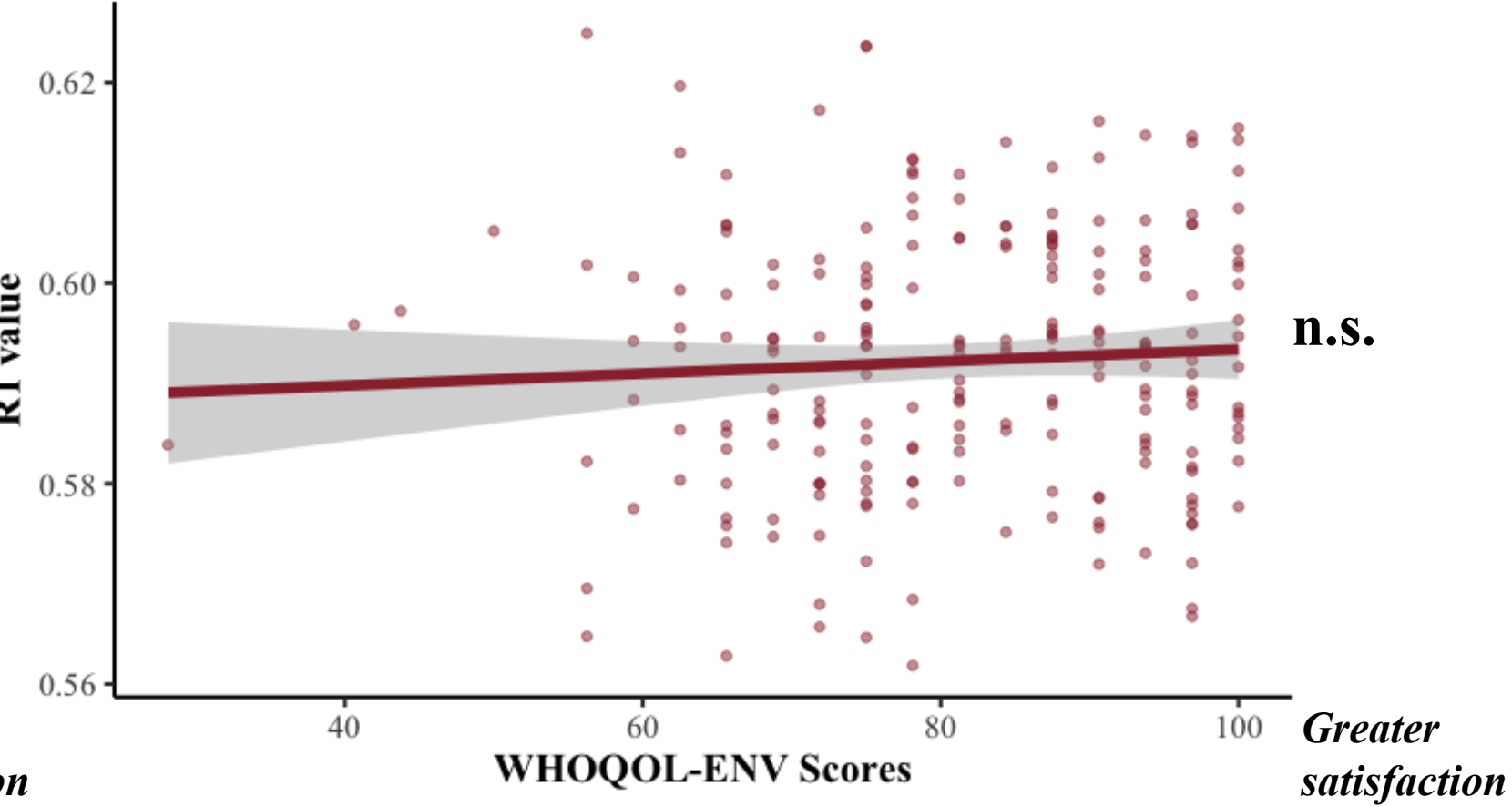
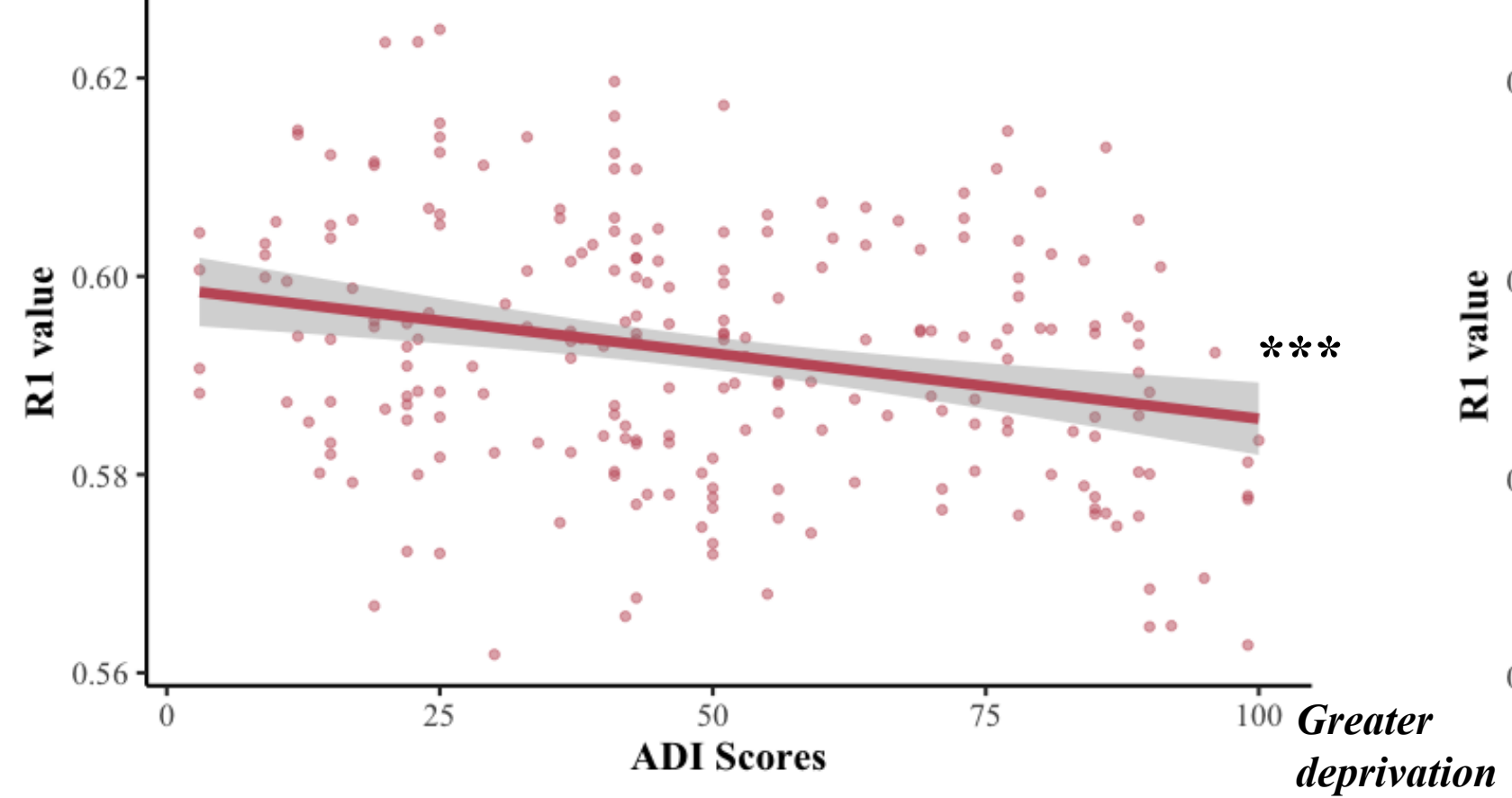
Distribution of Area Deprivation Scores (left) and WHOQOL-ENV scores (right).



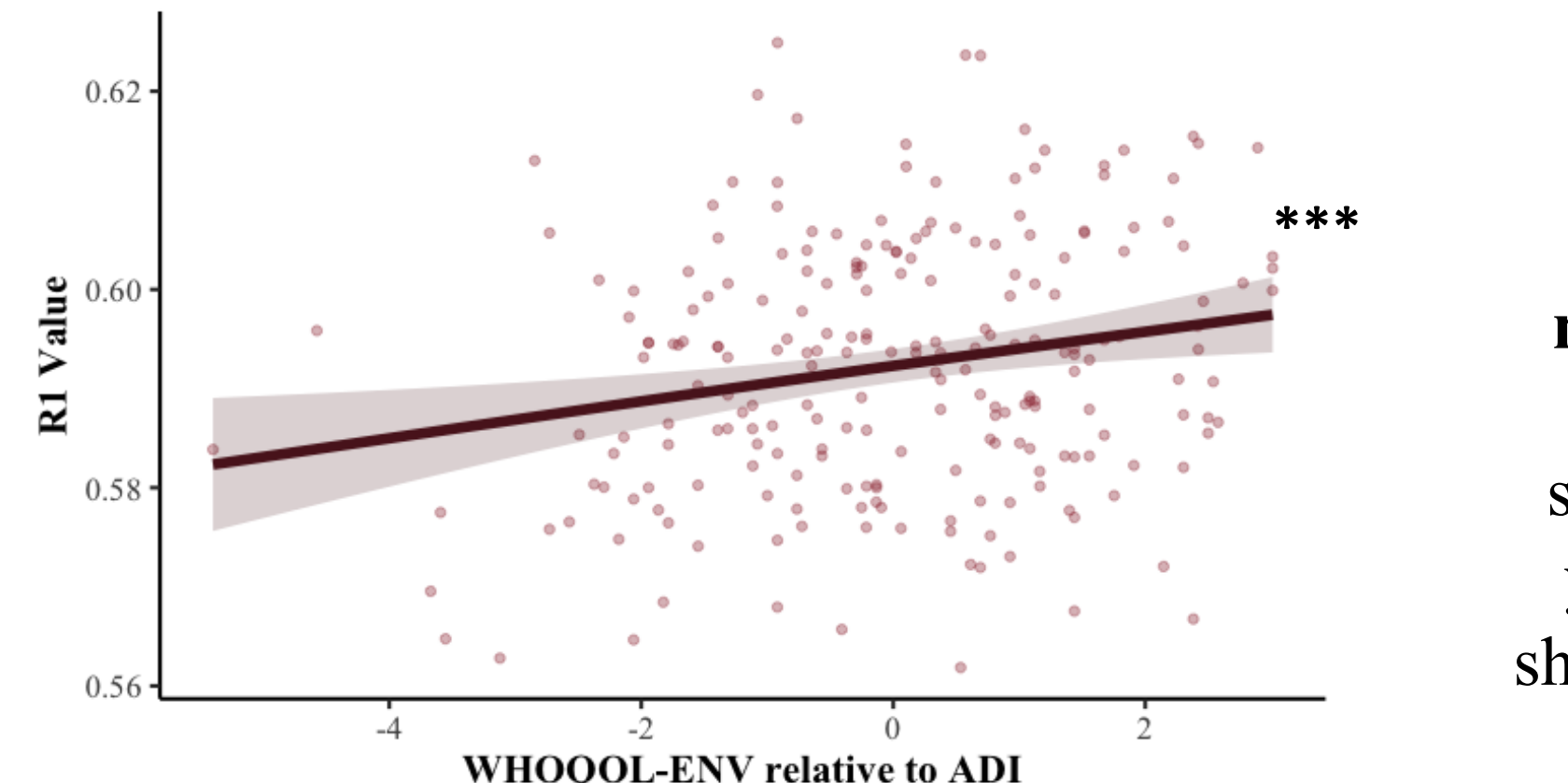
ADI and WHOQOL-ENV were significantly related to one another ($p = 0.038$). However, the effect size of this relationship was small, indicating that the ADI and WHOQOL-ENV capture distinct aspects of environmental variation.

Dominance analysis identified which WHOQOL-ENV items most strongly predicted objective neighborhood deprivation (ADI). Financial Resources was the strongest contributor, accounting for the largest share of explained variance. Smaller but meaningful contributions were observed for Health Services and Leisure Opportunities.

Objective Deprivation Predicts HPC-R1 and Effects of Environmental Perception



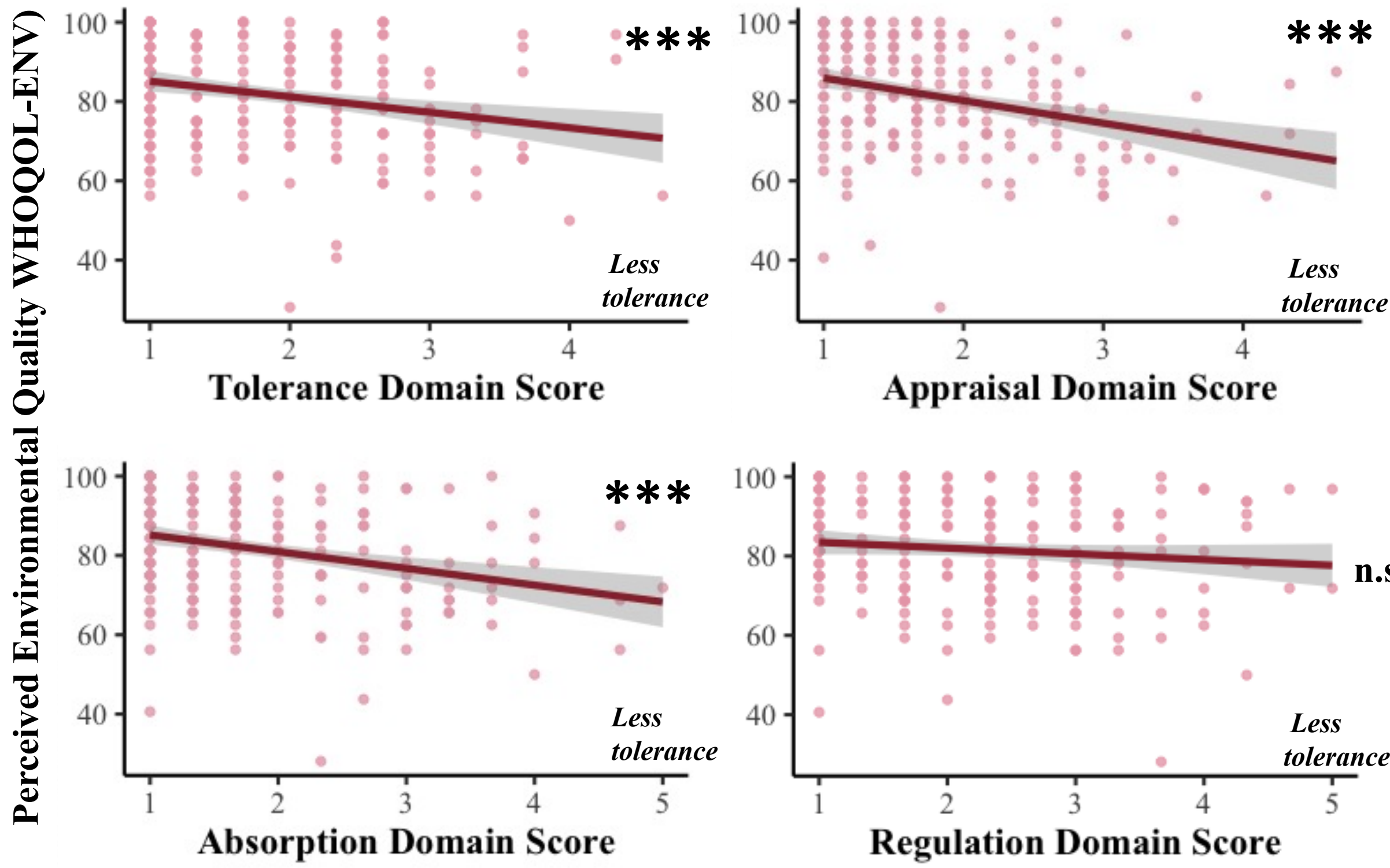
Controlling for age and sex, greater objective deprivation was statistically significant related to lower HPC R1 ($p = 0.000314$) and perceived environmental quality showed small, positive trend-level association with R1 ($p = 0.081$).



Youth who perceive their environment more positively than expected for their level of neighborhood deprivation show higher HPC R1.

When both WHOQOL-ENV and ADI were standardized (z-scored), controlling for age and sex, youth with higher WHOQOL(Env) relative to ADI showed higher hippocampal myelination ($p < 0.0001$).

Distress tolerance was associated with perceived environmental quality but does not moderate environment-R1 association



Higher WHOQOL(Env) scores were linked to lower distress tolerance across Absorption and Appraisal domains ($p < 0.001$) but not Regulation Domain ($p = 0.13$). However, none of the distress tolerance domains moderated the associations between ADI and R1 or WHOQOL(Env) and R1. Interaction terms were non-significant across all models.

Conclusions

- We show that R1 increased non-linearly in both HPC hemispheres, reflecting developmental increases in myelination through adolescence.
- R1 was more strongly associated with measures of objective neighborhood deprivation (ADI) compared to perceived environmental measures (WHOQOL-ENV), highlighting the utility of quantitative environmental measures in understanding neurobiological pathways.
- Although distress tolerance is a meaningful correlate of environmental perception, it does not alter the structural pathways of objective or subjective environmental quality to R1 microstructure, suggesting exploring other adaptive measures (e.g., behavioral) to understand individual-level resilience.
- Future work will examine whether distress tolerance may buffer in a developmentally-specific manner, such as it being critical for adulthood while peer and family support may be more important during adolescence.
- Overall, this work is informative to the critical need to address domestic polices in effort to reduce economic inequities and may provide insight into interventional approaches for at-risk youth.

Acknowledgements & Funding

Thank you to all participants!

Immense gratitude to all research assistants on data collection.

This research was supported by 5R01MH067924-19, the Staunton Farm Foundation, and CTSI UL17R001857.

References & Author Information



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